

Web Engineering 1 | WE1

Summary

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1. HYPERTEXT MARKUP LANGUAGE (HTML)

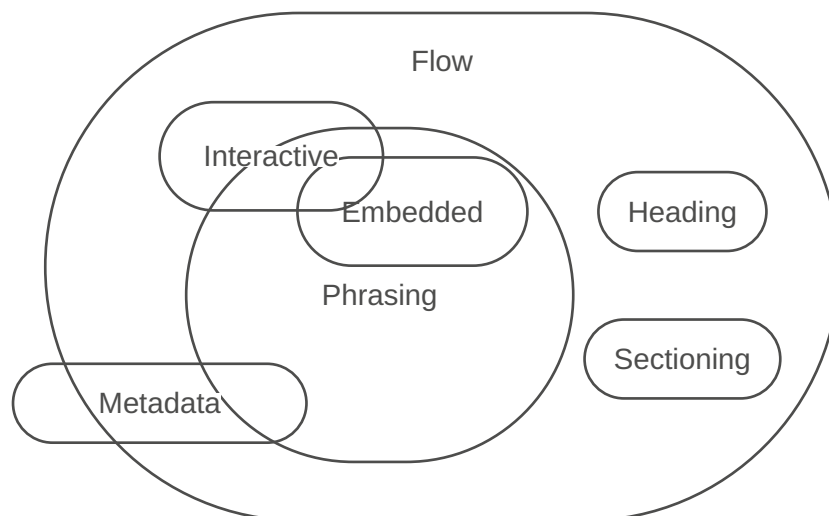
Structures the contents of a document, defines **what** is content and not **how** it looks.

<head> contains metadata, <body> contains the content.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <link rel="icon" href="ost-favicon-32x32.png">
  <title>Hello World</title>
</head>
<body>
  <h1>Hello World</h1>
  <p>Ich bin eine Testseite</p>
</body>
</html>
```

1.1. CONTENT CATEGORIES

Most HTML elements are a member of one or more content categories. The content categories are used to define the content model of elements, in other words, what each element can take as descendants.



Metadata: Elements belonging to the metadata content category modify the presentation or the behavior of the rest of the document, set up links to other documents, or convey other out-of-band information.

Flow: Flow content is a broad category that encompasses most elements that can go inside the <body> element, including heading elements, sectioning elements, phrasing elements, embedding elements, interactive elements, and form-related elements.

Sectioning: Creates a section in the current outline defining the scope of <header> and <footer> elements.

Heading: Defines the title of a section.

Phrasing: Refers to the text and the markup within a document. Sequences of phrasing content make up paragraphs.

Interactive: Includes elements that are specifically designed for user interaction.

```
graph TD
    Start([Start]) --> D1{Is it a major navigation block?}
    D1 -- YES --> Nav([<nav>])
    D1 -- NO --> D2{Does it make sense on its own?}
    D2 -- YES --> Article([<article>])
    D2 -- NO --> D3{Is it required to understand the curr. content?}
    D3 -- YES --> Aside([<aside>])
    D3 -- NO --> D4{Could you move it to an appendix?}
    D4 -- YES --> Figure([<figure>])
    D4 -- NO --> D5{Does it have any semantics?}
    D5 -- YES --> AppElem([Appropriate element])
    D5 -- NO --> Div([<div>])
```

The flowchart starts with a blue rounded rectangle labeled "Start". An arrow points down to a purple diamond decision box: "Is it a major navigation block?". If "YES", an arrow points left to an orange rounded rectangle: "<nav>". If "NO", an arrow points down to another purple diamond: "Does it make sense on its own?". If "YES", an arrow points left to an orange rounded rectangle: "<article>". If "NO", an arrow points down to a third purple diamond: "Is it required to understand the curr. content?". If "YES", an arrow points left to an orange rounded rectangle: "<aside>". If "NO", an arrow points down to a fourth purple diamond: "Could you move it to an appendix?". If "YES", an arrow points right to an orange rounded rectangle: "<figure>". If "NO", an arrow points right to a fifth purple diamond: "Does it have any semantics?". If "YES", an arrow points right to an orange rounded rectangle: "Appropriate element". If "NO", an arrow points up to an orange rounded rectangle: "<div>".

The diagram illustrates the layout of a basic HTML document. It is divided into five colored rectangular sections:

- <header>**: A blue section at the top.
- <nav>**: A purple section below the header.
- <main>**: A large red section below the navigation, occupying the left two-thirds of the width.
- <aside>**: A green section to the right of the main content, occupying the right one-third of the width.
- <footer>**: An orange section at the bottom, spanning the full width.

<h1> should only appear once on the website.

A self-contained part of the document which is independently placed or should be reused.

```
<article>
  <section id="introduction">
    <p>JS was a mistake</p>
  </section>
  <section id="content">
    <!-- very long rant -->
  </section>
  <section id="summary">
    <p>JS was a mistake</p>
  </section>
</article>
```

1.2.4. section

General separation and grouping of content.

1.2.5. img

alt attribute's value should provide a clear and concise text replacement for the image's content. It should not describe the presence of the image itself or the file name of the image. The alt attribute **must be specified**.

TODO:

<https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Elements>

2. CASCADING STYLESHEETS (CSS)

3. JAVASCRIPT (JS)

[why did you do this to us, brendan eich](#)

4. SEO AND ACCESSIBILITY